

Newton's Laws of Motion

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Newton's first law describes inertia, the second law connects force, mass, and acceleration ($F = m \times a$), and the third law describes equal and opposite interaction forces.
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Round 1 - Retrieve It

1. What is the big idea of Newton's first law?

2. Write Newton's second-law relationship.

3. What is the big idea of Newton's third law?

4. What happens to acceleration if the same force acts on a larger mass?

Round 2 - Sketch It

Draw a simple scene for each law: inertia, $F = m \times a$, and an action-reaction interaction. Label 1, 2, and 3.

Newton's Laws of Motion - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A 3 kg cart accelerates at 4 m/s^2 . What net force acts on it?

2. A passenger lurches forward when a car stops suddenly. Which law best explains this?

3. A swimmer pushes water backward and water pushes the swimmer forward. Which law?

Round 4 - Reason from Evidence

Students sometimes say third-law forces cancel each other. Explain why the pair does not cancel on one object.

Newton's Laws of Motion - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Newton's Laws of Motion in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect Newton's second law to net force. Which force value belongs in $F = m \times a$ when several forces act?

E**Extend**

Two carts get the same 20 N net force. One is 2 kg and one is 5 kg. Predict which accelerates more and calculate both accelerations.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Newton's Laws of Motion

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Objects resist changes in motion; motion changes only when a net external force acts.
2. $F = m \times a$.
3. For every interaction force, there is an equal-magnitude, opposite-direction force on the other object.
4. Acceleration decreases.

Sketch It - look for

Any accurate examples. Third-law pair must act on two different objects.

Apply It

1. 12 N.
2. First law/inertia.
3. Third law.

Reason from Evidence - look for

The equal and opposite forces act on different objects. Forces cancel only when forces on the same object sum to zero.

ACE - Extend look-for

2 kg cart: 10 m/s². 5 kg cart: 4 m/s². Smaller mass accelerates more with the same net force.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.